Dependency and Schedule Module Design

Student Project Planning and Progress Tracking System

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Status: DRAFT PENDING TEAM REVIEW. This is a targeted integration revision of the existing F3/F4 report after receiving the F1/F2 and F5/F6 designs. Original correct algorithms, numerical tables and all 17 planned cases are preserved. No application, customer approval or executed software-test result is claimed.

1 Scope and evidence

F3 Task Dependency Management and FR-09 are Should Have; F4 and FR-10/11/12 are Could Have [S2]. F5 FR-13/14 remain Must Have and F6 FR-15 Could Have. The Must dashboard blocker view depends on Should F3: keep basic progress/status/overdue information available if F3 is off, label blocker information unknown, and record the scope decision without upgrading URS priorities.

Requirement	Design allocation	Priority
FR-07 Create Task Dependencies	F3 validation and maintenance	Should
FR-08 Detect Invalid Circular Dependencies	F3 cycle detection before commit	Should
FR-09 Display Blocked Tasks and Reasons	F3 links; F5 derives and displays warnings	Should
FR-10 Calculate Estimated Project Duration	F4 ordering and forward calculation	Could
FR-11 Identify Schedule-Critical Activities	F4 backward calculation and zero float	Could
FR-12 Recalculate Schedule After Changes	F4 refresh coordination after valid saves	Could

DR-01 is a candidate finish-to-start calculation rule pending validation. DR-02 uses calendar days; DR-03 permits independent work in parallel; DR-04 permits manual updates despite blocker warnings; DR-05 leaves workload decisions to people [S2, p. 19].

Source verification: the named URS_REVISED PDF was recovered from the existing requirements repository history and is byte-identical to the 24-page URS used previously. The relevant FR/DR/NFR semantics therefore need no baseline-driven design change. The current revised requirements-analysis PDF also matches the prior attachment. Customer approval is not established by file recovery. See Section 10 and the review record.

2 Module decomposition

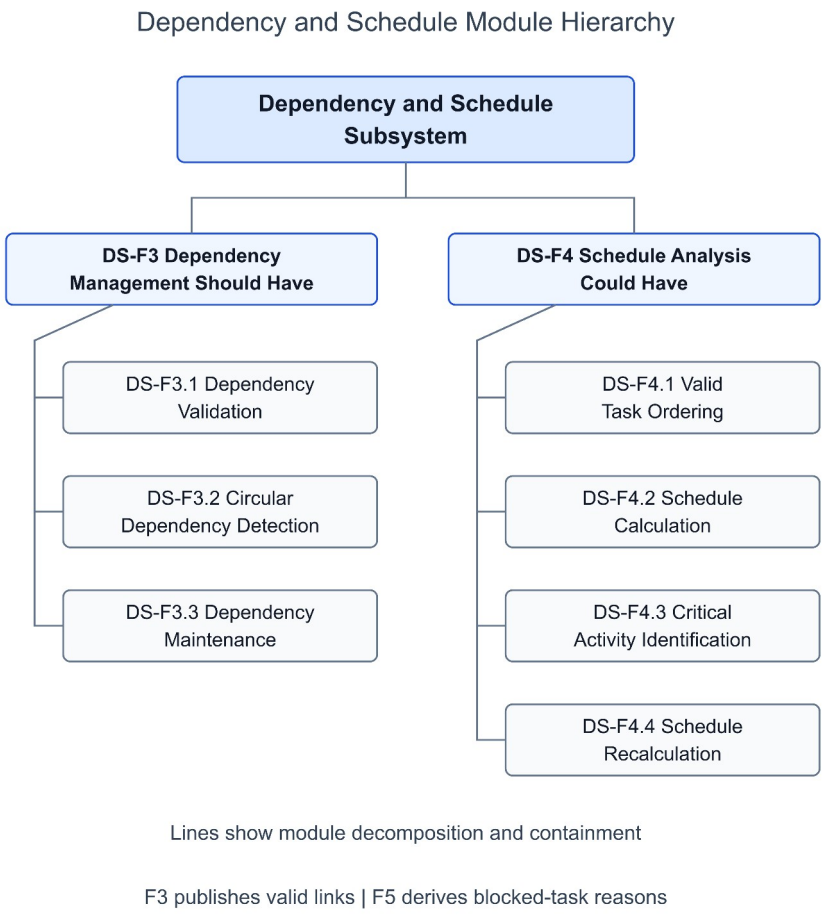


Figure1. Software decomposition with DS-F3.1-3.3 and DS-F4.1-4.4 identifiers. F3 supplies valid edges; only F5 derives Blocked. The matching ROUND2 draw.io remains editable. Names and editorial status belong to report metadata.

Module	Responsibility and hidden detail
DS-F3 Dependency Management	Own relationship validation and saved prerequisite links. Expose accepted links and understandable rejection reasons.
DS-F3.1 / DS-F3.2	Validation checks identities and rules; the graph helper only answers structural cycle questions. Keep authorization policy outside the graph algorithm.
DS-F3.3 Maintenance	Coordinate add/remove, candidate validation and atomic save. Supply prerequisite links for F5 without owning its progress display.
DS-F4.1 to DS-F4.4	Order a valid graph, calculate timings and criticality, and coordinate refresh. Hide traversal order and temporary timing arrays.

The children may be internal routines rather than separate deployable components. Ordering and cycle detection can share a private graph utility. Separate responsibilities improve cohesion without introducing microservices or an event broker.

3 Dependency validation and preservation

3 1 Proposed user interaction

The user opens a dependent task, selects a prerequisite and submits the relationship. We write u -> v to mean prerequisite u must finish before dependent v in the estimated plan. Membership checks occur for both viewing and editing; selected task names never grant access by themselves.

The use-case coordinator checks that the actor is a current project member and is authorized for dependency editing under the agreed policy. Exact leader/member dependency-edit rights remain OQ-06. The design passes an authorization decision into maintenance rather than assuming that all members can edit.

3 2 Candidate validation and save

Read the latest saved project snapshot. Resolve task references inside the actor's authorized project context before returning detailed validation. Unknown or inaccessible references receive the same user-facing message, "Invalid or inaccessible task." Do not reveal existence, private names, project titles or membership. Internally distinguish invalid IDs and cross-project references. For accessible tasks, check same-project membership, different IDs and duplicate links, then form a candidate graph without changing saved relationships.

For a proposed edge u -> v, search from v through successors. If u is reachable, adding the edge would close a cycle. A depth-first search can return the path used in the explanation. Alternatively, topological checking of the whole candidate graph fails when fewer than all tasks can be ordered. Both approaches take O(V + E) time on a small task graph. They do not require enumerating all possible paths.

SH-COORD requires expectedRevision for an existing-project mutation. Authorization, current-revision comparison, candidate cycle validation and source save form one protected commit decision. Advance revision once only after success, then send SavedChange. Two individually valid concurrent edits cannot jointly commit a cycle; a conflict requires a fresh snapshot and validation. Exact storage/locking technology remains provisional.

3 3 Required cycle example

Operation	Expected design behavior
Add A -> B	Accept if otherwise valid.
Add B -> C	Accept; saved graph is A -> B -> C.
Propose C -> A	Search from A reaches C through B. Reject the candidate.
Explain and preserve	Explain A -> B -> C -> A. Keep A -> B and B -> C, their revision and the previous valid schedule; retain user selections for correction.

3 4 Effect of an accepted relationship

After commit, F3 supplies the new prerequisite links. F5 combines them with the same saved task statuses: an unfinished task is blocked when at least one direct prerequisite is incomplete, and the warning names only that incomplete subset. Adding a link to an already completed prerequisite creates no blocker. Removing a link can remove a blocker. Completed dependent tasks are not counted as unfinished blocked work [S3, PDF p. 27].

F5 keeps the underlying status and permits authorized manual status updates. If retained, F4 refreshes the plan; a cycle rejection or failed save triggers neither a successful refresh nor a source-data replacement.

4 Schedule calculation and worked example

Precondition for a nonempty schedule: coherent ProjectSnapshot, complete valid F3 edges, acyclic graph and a finite positive durationDays for every task. Integer-only, mandatory-field and input-scale rules are not approved. NumericPolicy in Section4.4 preserves canonical precision; missing or invalid input makes F4 unavailable without automatically rejecting core F2 saving.

4 1 Ordering and forward calculation

A topological ordering puts every prerequisite before its dependent. Multiple zero-predecessor tasks are valid starts. A stable task-ID tie-break may make the display reproducible without changing calculated times. If ordering does not include every task, return an invalid-graph explanation instead of a partial schedule.

Day 0 is the planned start. For task v, ES(v) = 0 if it has no predecessors; otherwise ES(v) = max EF(u) over its predecessors. EF(v) = ES(v) + duration(v). Project duration T = max EF(v) over every task, including disconnected components. Intervals use elapsed boundaries, so [0, 2) lasts two calendar days.

4 2 Backward calculation and criticality

Use reverse topological order. Each terminal task has LF = T. Other tasks have LF(v) = min LS(w) over successors. LS(v) = LF(v) - duration(v). Total float = LS - ES. A task is mathematically critical at zero float; Section4.4 certifies that test through one NumericPolicy, never display rounding or unqualified binary equality. This common project-finish anchor handles separate branches and returns every critical activity, including tied critical paths. Deadlines are comparison targets, not backward-pass anchors.

Task	Days	ES	EF	LS	LF	Float	Critical
A Requirements	2	0	2	0	2	0	Yes
B Frontend	3	2	5	3	6	1	No
C Backend	4	2	6	2	6	0	Yes
D Testing	2	6	8	6	8	0	Yes

Dependencies: A -> B -> D and A -> C -> D. The branches last 7 and 8 days respectively. Estimated project duration is 8 days; the critical sequence is A -> C -> D. This is a design calculation, not an executed software test [S4, pp. 4-5].

4 3 Revision of the backend estimate

Task	Days	ES	EF	LS	LF	Float	Critical
A Requirements	2	0	2	0	2	0	Yes
B Frontend	3	2	5	5	8	3	No
C Backend	6	2	8	2	8	0	Yes
D Testing	2	8	10	8	10	0	Yes

When C changes from 4 to 6 days and saves validly, T becomes 10 days. Critical activities remain A, C and D; B now has 3 spare days. All independent activities may overlap, even with the same assignee.

4 4 Numeric precision and diagnostic policy

This proposed shared NumericPolicy remains subject to D-03/D-07 review. Retain canonical finite base10 input digits and do not round intermediate ES, EF, LS, LF, totalFloat or T. Prefer exact decimal or equivalent rational arithmetic: certified mathematical totalFloat=0 is critical; every genuine positive float is noncritical. Exact arithmetic makes this a justified equality, not an unexplained binary-float comparison. Presentation rounding is separate. No integer-only or fixed-scale business constraint is introduced.

A binary backend must use one central certified interval/error policy based on validated input conversion, supported range/scale, language/library rounding and graph propagation. Enclose addition/subtraction with outward rounding and max/min with predecessor/successor endpoint extrema. A strictly positive float interval certifies noncriticality. If it contains zero, or shows a negative/invariant conflict, recompute from canonical inputs with exact arithmetic or a higher-precision proof before classifying. Do not select an arbitrary fixed epsilon. Unresolved timing/criticality yields whole UNAVAILABLE/CALCULATION_FAILED, never a partial current critical set. Concrete limits, backend, bounds and display format are still pending.

After the feature, empty-task and dependency gates, validate every task duration. Missing, nonnumeric, nonfinite, zero or negative values yield whole UNAVAILABLE/INVALID_INPUT: durationDays, timings, criticalTaskIds and estimatedProjectFinish are null. For an already authorized coherent project, diagnostics may contain DurationInputDiagnostic {taskId, field="durationDays", issue=MISSING|NON_NUMERIC|NON_FINITE|NON_POSITIVE, message}. Report every detectable invalid field, not a partial valid schedule. Other result states use diagnostics=null. Denied/unknown project access returns the generic PublicError before analysis, with no private IDs, names, existence or values.

Analysis and diagnostics reference the attempted current sourceRevision; they neither increment revision nor publish SavedChange. Missing-duration core-save policy remains D-03: F4 failure alone does not reject an otherwise policy-valid F2 save or undo a successful source commit. Safe F5 raw counts, completion, assignments and overdue facts continue. With valid current F3/status inputs, MON-M3 still supplies current Blocked facts and direct incomplete reasons. Only warning types requiring absent F4 facts are unevaluated; enabled F6 can still assess independent overdue facts.

IT25 uses a standalone P25 revision7 snapshot: A2/B3 completed, C6 in progress, D2 not started, with diamond links. Relative T=10, critical A/C/D, B float3, raw2/1/1, completion50%, and D blocked only by C; enabled F6 warns for blocked-critical D. All derived sourceRevision values match7. asOfDate=2026-10-26 supplies read context, but plannedStartDate, projectDeadline and task deadlines are null; no absolute finish or project conflict is invented. IT26 includes exact decimal ties and a true1e-12 slack; IT27 contains synthetic invalid analysis fixtures, not a claim that F2 accepts malformed records. All27 integration scenarios remain NOT RUN.

PENDING D-04 recommends calendar days and Day0=plannedStartDate. If explicitly selected for a conditional scenario or later approved, estimatedProjectFinish=plannedStartDate+ceil(T), using the full canonical unrounded NumericPolicy-resolved T. Resolve any binary uncertainty crossing an integer boundary before ceil. Unknown start uses MISSING_PLANNED_START; unconfirmed mapping uses DATE_MAPPING_UNRESOLVED and finish=null. Relative analysis can remain VALID. Task/project dates stay separate, strict > comparisons exclude equality, and missing dates are unevaluated rather than safe. IT12/16/21/22 are conditional on this policy; the chat timezone grants no approval.

5 Dependency interface contracts

These are conceptual design contracts, not source code or implemented APIs. Public mutations use MutationResult<T>; derived views use availability and sourceRevision. For inaccessible project/task references return the generic public message without private names, membership, saved values or existence. Authorized self-link/duplicate/cycle explanations remain detailed.

addDependency DS-F3.3

Input: actorId (UserId), projectId, prerequisiteId, dependentId and expectedRevision.

Output: MutationResult<DependencyEdge> with committed revision and SavedChange, or a public failure with no successful change or private saved payload. Cycle paths are returned only within an authorized context.

Contract: Require membership and policy permission; validate the candidate against the latest graph before atomic save. On rejection or save failure keep saved source data and current valid results unchanged. Notify downstream refresh only after success.

removeDependency DS-F3.3

Input: actorId (UserId), projectId, prerequisiteId, dependentId and expectedRevision.

Output: MutationResult with committed revision and SavedChange after removal, or public permission/input/revision/save failure. Missing accessible links return an authorized invalid-input explanation without a mutation.

Contract: Apply the same membership and operation policy. Resolve inaccessible tasks generically before relationship details. Removal cannot introduce a cycle but must pass expectedRevision and atomic-save rules. Only a successful source commit triggers F5 and retained F4 refresh.

validateDependency DS-F3.1

Input: read-only authorized project snapshot and proposed relationship.

Output: internal valid/invalid outcome with invalid-ID, self-link, duplicate, cross-project or cycle reason; the coordinator maps it to a permitted user-visible message.

Contract: No writes. Validate graph rules independently of UI choices. Membership and authorization are coordinator preconditions, not inferred from IDs. Map unknown or inaccessible references to the same generic explanation, "Invalid or inaccessible task." Never return their names, project titles, membership details or existence. Preserve useful explanations for accessible project relationships. Preflight success does not replace revalidation at commit.

detectCycle DS-F3.2

Input: candidate directed graph with verified task endpoints.

Output: hasCycle plus a task-ID path when detected, or noCycle.

Contract: Private graph operation; no persistence, authorization decisions, task reassignment or status changes. Validate all components when using a full DFS or topological check. A self-link is invalid before this helper is normally reached.

5 1 Information hiding

Only maintenance writes dependency records. Clients do not receive mutable adjacency lists or temporary DFS markers. Change the internal cycle algorithm without changing add/remove outcomes. Hide storage errors behind a stable failure contract while retaining enough detail for the user to correct the request.

Suggested result reasons are conceptual: invalid relationship, circular relationship, permission denied, changed plan and save failed. Exact names and UI wording will be agreed during detailed design.

6 Schedule interfaces and data exchange

calculateSchedule DS-F4.1 and DS-F4.2

Input: one coherent ProjectSnapshot with revision, tasks, dependencyView and optional plannedStartDate; retained F4 configuration. projectDeadline/taskDeadline do not constrain CPM.

Output: ScheduleResultView with availability/sourceRevision, nullable schedule facts, scoped reasonCode/finishReasonCode and diagnostics. Section12 lists the legal state/reason pairs; Section4.4 limits diagnostics to authorized INVALID_INPUT, otherwise null.

Contract: No source writes. Gates: F4 off -> NOT_ENABLED; enabled empty -> NO_TASKS; otherwise require complete valid current dependencyView (sourceRevision=snapshot.revision). Missing/untrusted input -> UNAVAILABLE/DEPENDENCY_INPUT_UNAVAILABLE; never treat edges=null as edges=[]. Invalid schedule inputs -> UNAVAILABLE/INVALID_INPUT.

identifyCriticalActivities DS-F4.3

Input: VALID ScheduleResultView and current snapshot revision; sourceRevision must match.

Output: complete NumericPolicy-certified criticalTaskIds at current sourceRevision. Unavailable/noncurrent facts are null. F5/F6 apply no local epsilon.

Contract: No need to enumerate all critical paths. Reject unavailable or mismatched results. Do not infer business priority, actual remaining effort, resource feasibility or task ownership.

recalculateSchedule DS-F4.4

Input: SavedChange and the latest coherent ProjectSnapshot; optional F4 configuration.

Output: ScheduleResultView at current sourceRevision, or NOT_ENABLED/NO_TASKS/UNAVAILABLE with a specific reason and null noncurrent payload.

Contract: No source writes. Refresh after successful source save; recompute changed schedule inputs. Reuse requires Section11 proof and current absolute-date mapping. Rejected writes publish no result; later analysis failure preserves the saved source.

6 1 Input required from Task Management

Data	Purpose
Project and access	projectId, authorized member/role context, plannedStartDate, projectDeadline, coherent revision. IDs are opaque; actorId UserId differs from assigneeId MemberId.
Task records	TaskSnapshot: taskId, projectId, name, assigneeId|null, durationDays, taskDeadline|null, status=NOT_STARTED|IN_PROGRESS|COMPLETED. F2 owns these source values.
Relationship records	F3 DependencyResultView: availability/sourceRevision/edges/reasonCode. Off uses FEATURE_NOT_ENABLED; UNAVAILABLE uses DEPENDENCY_INPUT_UNAVAILABLE, INVALID_INPUT or STALE_RESULT, not CALCULATION_FAILED (Section12). Known edges=[] differ from unknown edges=null.

6 2 Output for Progress and Risk Monitoring

F5 alone derives progress and direct incomplete blockers from F2 statuses/F3 links. F4 supplies current schedule facts. MON-M5 overdue uses taskDeadline/asOfDate independently of F4/F6. F6 composes retained warnings behind current sourceRevision/availability gates; it never runs CPM.

7 Refresh rules and planning assumptions

Saved change or event	Required derived refresh
Add/delete task; edit duration; add/remove dependency	Recompute retained F4 in full; refresh affected F5/F6 views. Task deletion and link cleanup must form one agreed valid change.
Edit plannedStartDate	Revalidate source revision; reuse relative timings only for unchanged graph/durations, remap absolute finish and refresh project comparison.
Edit projectDeadline or taskDeadline	Refresh respective project comparison or MON-M5 basic overdue facts and optional F6 warnings; never fit CPM to a target date.
Edit task status	Refresh F5 progress/blockers and relevant F6 warnings. Do not infer shorter durations or remaining effort.
Edit assignee or task name	Refresh assignments or labels. Ownership alone does not change the schedule.
Change asOfDate	Refresh MON-M5 basic overdue facts and optional F6 warnings; no source revision increment or inherent CPM recalculation.
Invalid edit or failed save	Preserve prior saved plan and its valid derived results; show unsaved input and the failure.

One atomic source change advances revision once. SH-COORD reads project, tasks, memberships and complete F3 edges coherently. DependencyResultView.sourceRevision equals snapshot.revision. After notes/roles change, reauthorize and prove task universe/validated edges unchanged, or re-read, before current publication. Task/link changes need fresh graph validation. Failed proof gives dependency UNAVAILABLE/STALE_RESULT with edges=null. Schedule reuse also proves task IDs, graph and durations unchanged and remaps absolute finish. Failed proof gives schedule UNAVAILABLE/STALE_RESULT. Section11 defines this shared protocol; no extra counter or implemented concurrency mechanism is claimed.

A valid source save remains committed if analysis fails. Invalid duration gives UNAVAILABLE/INVALID_INPUT; unresolved precision or calculation failure gives UNAVAILABLE/CALCULATION_FAILED, with null schedule facts. Old results stay historical. Reliable F5 counts, assignments, overdue and current F3/status blockers continue. Section4.4 defines authorized diagnostics and separate F2 save policy.

7 1 Assumptions and boundaries

The pending D-04 calendar-day candidate includes weekends and holidays. Finish-to-start timing describes the plan without locking real work. Independent tasks run in parallel; a join waits for its latest predecessor. The group handles resource conflicts/reassignment. Entered estimates do not guarantee completion.

No automatic resource leveling, reassignment, working-day calendar, AI planning or optimization is proposed. A task already IN_PROGRESS may still carry a blocker warning. Whether completing a task before its prerequisites requires an extra confirmation is unresolved, and actual early starts do not automatically change the candidate calculation rule.

7 2 Proportional implementation plan

Proposed architecture: ordinary modules in one application, with a shared project snapshot and a small graph utility. Full topological, forward and backward passes take O(V + E) time and O(V + E) space. The URS typical 15-40 tasks is an example workload, not an approved limit or measured performance claim. Avoid incremental graph caches and distributed analysis until evidence requires them.

The technology stack, storage engine, UI framework and exact save mechanism remain undecided. The URS 1-2 second refresh expectation requires agreed devices, loads and measurement boundaries before any performance acceptance claim [S2, pp. 18, 24].

8 Proposed module verification cases

All cases below are design test plans. No application, automated module test or system acceptance test was executed for this deliverable. Numerical outputs are analytical oracles for later implementation checks.

ID and case	Setup or action	Expected result
T01 Valid relationship	Authorized editor adds A -> B within a project.	One saved link; F5 names A as blocker only while A and B are unfinished.
T02 Direct cycle	A -> B saved; propose B -> A.	Reject; explain A -> B -> A; graph, revision and valid schedule unchanged.
T03 Indirect cycle	A -> B and B -> C saved; propose C -> A.	Reject final link and explain the complete cycle; preserve earlier links.
T04 Invalid endpoints	Self-link, duplicate, unknown ID or cross-project endpoint.	Reject without partial save. Unknown/inaccessible references use the same generic message; disclose no names, membership or existence. Detailed reasons are limited to authorized project contexts.
T05 Multiple starts	A=2 and B=3 are independent; both precede D=1.	ES(A)=ES(B)=0; ES(D)=3; T=4; critical B,D.
T06 Parallel branches	A=2, B=3, C=4, D=2; required diamond.	T=8; critical A,C,D; B float=1.
T07 Equal critical branches	Same diamond, B=C=4.	T=8; all A,B,C,D critical, not only one branch.
T08 Changed critical branch	Same diamond, B=5 and C=4.	T=9; critical A,B,D; C float=1.
T09 Duration refresh	Save C=6 in the baseline diamond.	Automatic retained-F4 refresh: T=10; A,C,D critical; B float=3.
T10 Invalid duration	Missing, zero, negative, nonnumeric or nonfinite duration.	UNAVAILABLE/INVALID_INPUT with no partial schedule or zero-duration guess. Authorized DurationInputDiagnostic identifies taskId/durationDays and the applicable issue; core task-save acceptance remains the separate pending F2/D-03 policy.

9 Further cases and review decisions

ID and case	Setup or action	Expected result
T11 Empty graph	Project contains no tasks.	Proposed NO_TASKS, no estimate/critical set and no fabricated finish. Customer-facing empty behavior pending.
T12 Human workload	Two independent tasks of 3 and 4 days share one owner.	Both ES=0; T=4. Owner and ordering unchanged; people resolve conflict.
T13 Blocked manual update	B depends on incomplete A; update B to IN_PROGRESS.	Authorized update allowed; B keeps its status and names A as incomplete. No duration inference.
T14 Permission and save	Nonmember; unauthorized editor; storage failure.	Deny or explain failure; preserve source plan and entered values. Test approved role matrix once agreed.
T15 Concurrent edits	Two clients propose reverse links from one revision.	At most one accepted without a fresh cycle check; second retries against current graph.
T16 Removal and deletion	Remove last blocker; separately delete a task with links.	Successful link removal refreshes warnings. Deletion follows the later agreed reject/cascade policy atomically.
T17 Components and freshness	Independent terminals A=2 and B=5; analysis failure after valid edit.	T=5 and only B critical; never label old timing as current after a newer source save.

9 1 Decisions needed before detailed design

Confirm dependency-edit rights and leader agreement; D-03 mandatory fields, supplied-estimate validation and referenced deletion; D-03/D-07 NumericPolicy, supported input limits and implementation backend; D-04 calendar-day plannedStartDate+ceil(T) mapping and project timezone; status exceptions, empty presentation, capacity/recovery acceptance and retained optional F4/F6 scope. These are recommendations or open decisions, not customer approval.

The actual F1/F2 14-page and F5/F6 31-page designs have now been received, compared and incorporated into this ROUND2 draft. Shared contracts are editorial proposals awaiting team/customer decisions; receipt/static review does not establish agreement or implementation readiness.

9 2 Subcases for later test planning

Keep T01-T17 as the original scenarios. The combined cases below should become separately identifiable checks during detailed design and implementation testing. These identifiers are a plan; no application tests were executed.

Original case	Proposed independent subcases
T01	T01-A valid link save; T01-B blocker derivation before and after prerequisite completion.
T04	T04-A self-link; B duplicate; C unknown ID; D cross-project reference; E inaccessible reference and no information disclosure. Compare C/D/E user messages under inaccessible contexts.
T10	T10-A missing duration; B zero; C negative; D nonnumeric; E nonfinite. Task-save policy is separately agreed with F2.
T13	T13-A authorized manual update; B retained blocker warning; C unchanged entered durations.
T14	T14-A nonmember access; B member without edit permission; C source-save failure.
T16	T16-A remove dependency; T16-B delete a task with linked dependencies under the agreed lifecycle policy.
T17	T17-A independent terminal tasks; T17-B stale result handling after a valid save and calculation failure.

10 Source register and design status

[S1] SE_Chapter6-Structured Software Design-2026.10.22.pdf, 105 PDF pages. Stored at 00_Source_Materials/Course_Lectures/SE_Chapter6-Structured Software Design-2026.10.22.pdf. Page references use physical PDF pages.

[S2] Student_Project_Planning_and_Progress_Tracking_System_URS_REVISED.pdf, 24 pages, dated 10 October 2026. Recovered from Requirements_Specification Git commit 148981bced19f2dc2e16ca4d326e2fc1e5f78b60, path outputs/Student_Project_Planning_and_Progress_Tracking_System_URS_REVISED.pdf, into 00_Source_Materials/Task2_URS/. Its SHA-256 matches the URS(1).pdf used in the original design. The revised artifact is verified; customer approval is not inferred. Relevant pp. 12-14, 18-21, 24.

[S3] Student_Project_Planning_and_Progress_Tracking_System_Requirements_Analysis_REVISED.pdf, 36 pages, dated 10 October 2026. Available at Requirements_Analysis/outputs/. Its SHA-256 matches the previously supplied Requirements_Analysis(1).pdf; see the review evidence manifest. Relevant PDF pp. 7, 19-21, 26-29, 33-35.

[S4] Student_Project_Planning_and_Progress_Tracking_System_Proposal(1).pdf, 8 pages. Stored at 00_Source_Materials/Task1_Proposal/Student_Project_Planning_and_Progress_Tracking_System_Proposal(1).pdf. The separately requested Proposal_FINAL filename was not found. Relevant pp. 3-5 planning rules and worked example; p. 8 functional ownership.

[S5] Current Group9_Scaffold documentation and dated22 October review record. Pre-round Git HEAD is 10e23fa929e23aa2816f370f4bd1be88d4171831; ROUND2 documentation edits are uncommitted. Historic scaffold notes are background, not authority over URS.

10 1 Use of Chapter 6

We apply modular decomposition, high cohesion, low coupling and information hiding to the object-oriented requirements models. The hierarchy is a functional module proposal; it is not a mechanical DFD-to-structure-chart transformation. Task 3 used object-oriented requirements analysis [S3, PDF pp. 2, 7]. Lecture page-specific design references are recorded in the integration document.

10 2 Handoff package

02_Dependency_Schedule_Module_Design_ROUND2.docx/.pdf and 02_Dependency_Schedule_Hierarchy_ROUND2.drawio/.png are this round's artifacts. Original REVIEWED files remain intact. All seven operations, T01-T17 and subcases remain conceptual/planned. The shared contract and integration matrix are independent attachments.

[S6] Zhang Weiyi, 01_Project_Task_Module_Design(1).pdf, 14 pages, Version 1.0, Week2026-10-22. Original retained in Individual_Designs/ZhangWeiyi/Sources/. Received and compared for this round; source has no corresponding editable DOCX here.

[S7] Zhang Yuyang, 03_Progress_Risk_Module_Design(1).pdf, 31 pages, Version 1.0,25 October 2026. Original retained in Individual_Designs/ZhangYuyang/Sources/. Received and compared for this round; editorial corrections do not imply owner approval.

11 Shared revision and save protocol

One proposed project revision protocol applies across all three areas. SH-COORD is a shared responsibility realized by MG-M0/MON-M0, not an extra service. MG-M2 supplies the common project membership/edit policy.

Stage	Design invariant
Read	Authenticate/authorize the project context, then read one coherent ProjectSnapshot of F1/F2 records and F3 links at revision r. Do not merge independent current reads without consistency proof.
Mutate	Every existing-project write supplies expectedRevision=r. The protected commit checks authorization, current revision and all invariants; dependency cycle validation belongs to that same decision.
Commit	Create project at revision 1. Each successful atomic source change advances the revision once; a batch such as member departure/unassignment is one change. Rejected/failed requests do not increment or publish success.
Refresh	SavedChange names projectId, revision, changeKind, affectedTaskIds and changedFields. Use ordinary coordinator calls after success, without introducing a broker.
Derive	Use sourceRevision to reference that same counter. F5/F6 only use facts matching the current snapshot. sourceRevision is not an independent version field.
Reuse	Prove task IDs, graph and durations unchanged before reusing relative timings. Remap estimatedProjectFinish if plannedStartDate changes; refresh labels/comparisons and republish at current sourceRevision. Otherwise mark STALE_RESULT unavailable.

Concurrent reverse-edge requests from revision r cannot both commit: one succeeds or conflicts; the next attempt must read and revalidate against the resulting graph. Storage/locking technology and measurable performance remain detailed-design decisions. A change of asOfDate is a read and does not advance revision.

12 Shared availability and degraded views

Availability	Meaning and payload
VALID	Nonempty valid schedule at current sourceRevision: durationDays, timings and complete NumericPolicy-certified criticalTaskIds. Missing plannedStartDate -> finish=null/MISSING_PLANNED_START; unresolved mapping -> finish=null/DATE_MAPPING_UNRESOLVED. Reliable relative analysis remains VALID.
NO_TASKS	Enabled analysis with no tasks. Schedule durationDays/criticalTaskIds/estimatedProjectFinish are null. Progress totalTasks=0, completedTasks=0, percentage=null. Do not fabricate a delivery estimate or0%.
NOT_ENABLED	The optional feature is not retained/enabled. Payload is null; reasonCode=FEATURE_NOT_ENABLED. Distinct from a known empty result.
UNAVAILABLE	Use reasonCode INVALID_INPUT, CALCULATION_FAILED, STALE_RESULT or DEPENDENCY_INPUT_UNAVAILABLE. Current schedule facts are null; no stale highlights or finish warnings are presented as current.

DependencyResultView uses NOT_ENABLED/FEATURE_NOT_ENABLED with edges=null when F3 is off; enabled empty tasks -> NO_TASKS with edges=[]; enabled nonempty tasks with complete valid current edges -> VALID, including edges=[] for known no prerequisites. UNAVAILABLE carries DEPENDENCY_INPUT_UNAVAILABLE, INVALID_INPUT or STALE_RESULT, with edges=null. F4 applies its own off/empty gates first, then maps required unknown dependencies to UNAVAILABLE/DEPENDENCY_INPUT_UNAVAILABLE. F5 never reports unknown blockers as none.

A schedule-only failure preserves reliable progress, raw statuses, assignments and MON-M5 basic overdue facts. If F6 is enabled and taskDeadline/asOfDate facts are valid, OVERDUE_TASK remains independently evaluable; missing current criticalTaskIds or estimatedProjectFinish leaves BLOCKED_CRITICAL_TASK and PROJECT_DEADLINE_CONFLICT unevaluated. evaluationByType distinguishes unknown from no risk; isCritical remains nullable. F6 off keeps FR-15 NOT_ENABLED without removing the FR-13 overdue list.

Only unavailable/unsafe source reads or inaccessible project contexts justify a top-level failure. Unknown/inaccessible project/task references receive the generic public messages; detailed authorized graph errors remain useful. NFR-05 is Reliability: preserve entered input and valid saved records, reject invalid edits without replacing the valid plan, and plan recoverability after temporary unavailability.

13 Canonical contracts and review handoff

Record or rule	Shared shape and boundary
TaskStatus and display	NOT_STARTED | IN_PROGRESS | COMPLETED. rawStatus is independent from nullable isBlocked/isCritical on TaskDisplayView; BLOCKED never becomes a stored status.
ScheduleResultView	availability; sourceRevision; reasonCode?; durationDays|null; timings|null; criticalTaskIds|null; estimatedProjectFinish|null; finishReasonCode?; diagnostics:DurationInputDiagnostic[]|null. Diagnostics are authorized INVALID_INPUT validation details only; other states use null. TaskTiming retains taskId/ES/EF/LS/LF/totalFloat.
MutationResult	ok=true, projectId, revision, value, SavedChange only after atomic success; otherwise PublicError with no private saved payload or success fact. expectedRevision is mandatory for existing-project mutations.
Lifecycles	Member departure clears assignments and active access, preserves tasks/edges/notes/statuses and historical author references. Linked task deletion is blocked under the unresolved reference-policy proposal; no CANCELLED status or automatic reassignment.
Date candidates	PENDING D-04 candidate: calendar-day Day0=plannedStartDate; estimatedProjectFinish=plannedStartDate+ceil(canonical unrounded NumericPolicy-resolved T). Unfinished overdue iff asOfDate>known taskDeadline; project risk iff valid finish>known projectDeadline. Equality triggers neither. Missing/unconfirmed dates are unevaluated; project timezone, input limits and approval remain open.
Priority dependency	FR-13 base -> MON-M0 / MON-M1 / MON-M2 / MON-M4 / MON-M5 + UI. Blocked subview -> MON-M3, dependent on Should FR-09/F3; D-01 remains pending. F3 off/unavailable -> unknown blocker capability while base dashboard continues; fallback is not full acceptance.

The five ROUND2 attachments define shared contracts, formal traceability, conflict/decision recommendations,27 NOT RUN integration scenarios and change/evidence records. IT25 links valid relative CPM to current F5/F6 facts; IT26 checks NumericPolicy; IT27 checks authorized duration diagnostics and safe degradation. Correct original graph algorithms, two numerical tables and T01-T17 remain. Detailed implementation, source-save policy and all D-01..D-10 decisions remain pending.

Source-to-revision mapping: Sections1/3/5-7/9-10 and the hierarchy labels changed for shared contracts; Section4 calculations and all 17 verification scenarios remain preserved. Sections11-13 make the version, availability and lifecycle rules explicit. Source hashes and detailed changes are in Change_Log_and_Evidence_Index_ROUND2.md.